

Introduction to Causal Inference for Software Engineering

A 90-minute hands-on tutorial

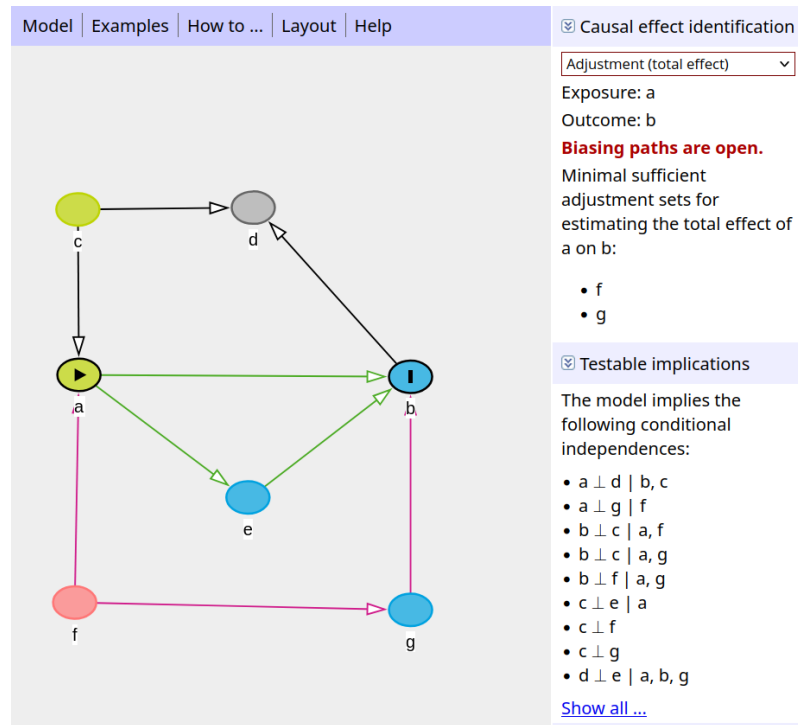
Julien Siebert · Julian Frattini · Hans-Martin Heyn · Roberto Pietrantuono

Section 4

Exercises: Identification, adjustment sets and conditional independence

A guides tour through DAGitty

- DAGitty is a browser-based tool to create and analyse DAGs
- It helps causal inference by automatically identifying necessary adjustment sets.
- Based on the graphical causal model, it provides ***conditional independence criteria*** that can be used to test models.
- Access it here: <https://dagitty.net/>



The screenshot shows the DAGitty web application interface. At the top, there are navigation tabs: Model, Examples, How to ..., Layout, and Help. The main area displays a directed acyclic graph (DAG) with nodes labeled a through g. Node 'a' is highlighted with a yellow circle and a play button icon. Node 'b' is highlighted with a blue circle and an 'I' icon. The graph shows causal relationships: 'c' points to 'd' and 'a'; 'a' points to 'e'; 'e' points to 'b'; 'f' points to 'g'; and 'g' points to 'b'. There are also unidirectional arrows from 'c' to 'a' and 'd' to 'b'. The right sidebar contains two sections: 'Causal effect identification' and 'Testable implications'. In the 'Causal effect identification' section, 'Adjustment (total effect)' is selected in a dropdown menu. Below this, it specifies 'Exposure: a' and 'Outcome: b'. A red warning message states 'Biasing paths are open.' followed by 'Minimal sufficient adjustment sets for estimating the total effect of a on b:' and a list containing 'f' and 'g'. The 'Testable implications' section states 'The model implies the following conditional independences:' followed by a list of 10 conditional independence statements. A 'Show all ...' link is at the bottom of the sidebar.

Model | Examples | How to ... | Layout | Help

☒ Causal effect identification

Adjustment (total effect) ▼

Exposure: a
Outcome: b

Biasing paths are open.
Minimal sufficient adjustment sets for estimating the total effect of a on b:

- f
- g

☒ Testable implications

The model implies the following conditional independences:

- $a \perp d \mid b, c$
- $a \perp g \mid f$
- $b \perp c \mid a, f$
- $b \perp c \mid a, g$
- $b \perp f \mid a, g$
- $c \perp e \mid a$
- $c \perp f$
- $c \perp g$
- $d \perp e \mid a, b, g$

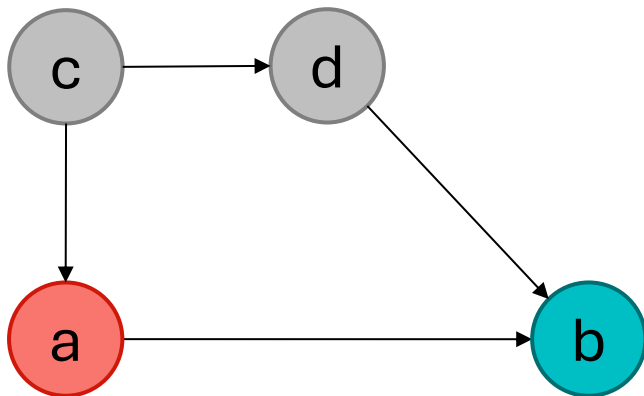
[Show all ...](#)

Textor et al., [Robust causal inference using directed acyclic graphs: the R package 'dagitty'](#). *International Journal of Epidemiology* 45(6):1887-1894, 2016.

Live Demo

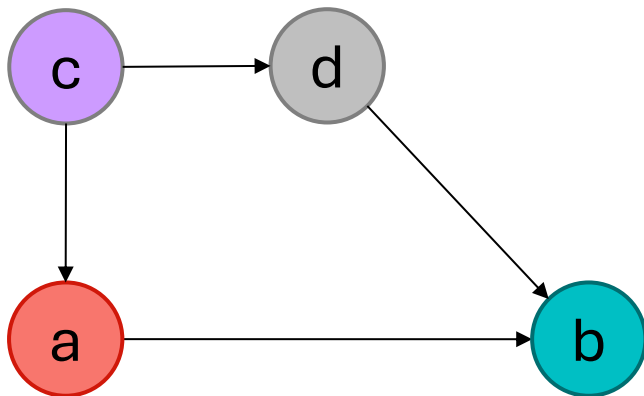
Problem 1

- Determine the **adjustment set** of variables to block any backdoor causal path for the estimation of the direct effect of a on b.

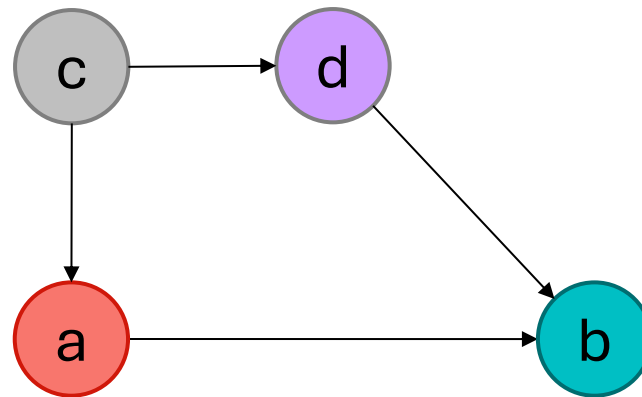


Problem 1 Solution <https://dagitty.net/mDymjnHV9>

- Determine the **adjustment set** of variables to block any backdoor causal path for the estimation of the direct effect of a on b.



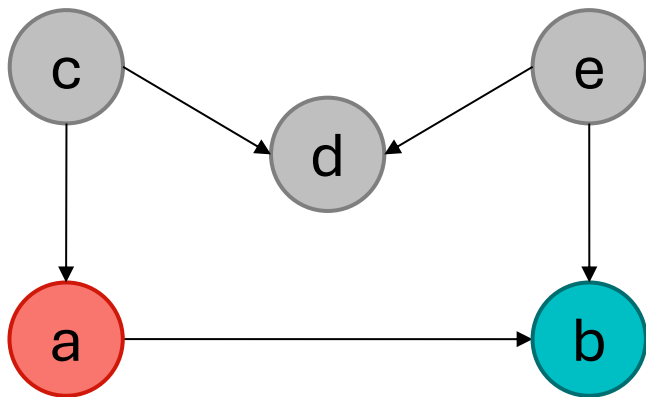
Adjustment set = {c}



Adjustment set = {d}

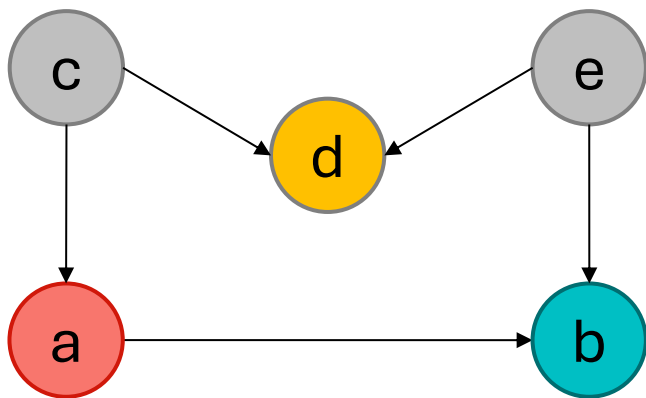
Problem 2

- Determine the **adjustment set** of variables to block any backdoor causal path for the estimation of the direct effect of a on b.



Problem 2 Solution <https://dagitty.net/mPxsokVRw>

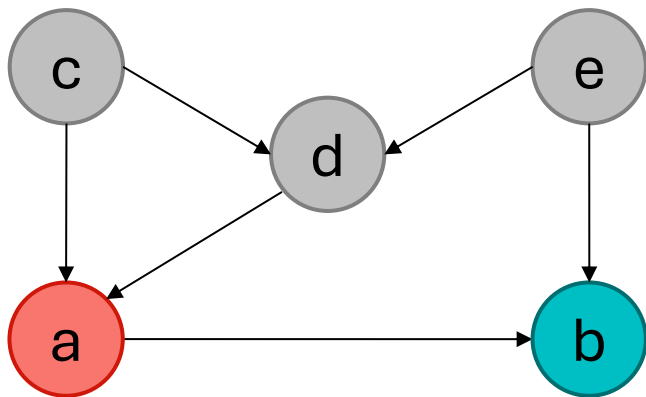
- Determine the **adjustment set** of variables to block any backdoor causal path for the estimation of the direct effect of a on b.



Adjustment set = {}

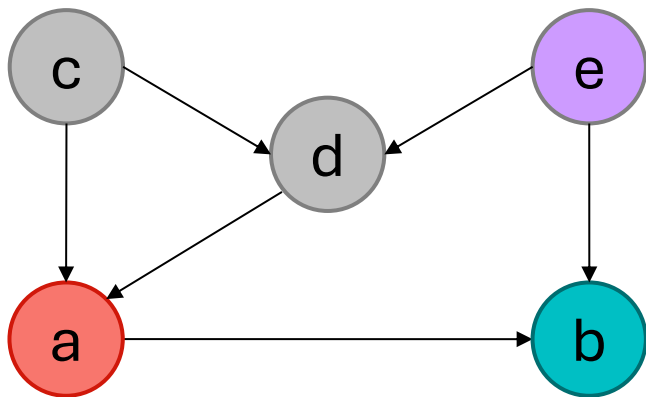
Problem 3

- Determine the **adjustment set** of variables to block any backdoor causal path for the estimation of the direct effect of a on b.



Problem 3 Solution <https://dagitty.net/mvzunz7xx>

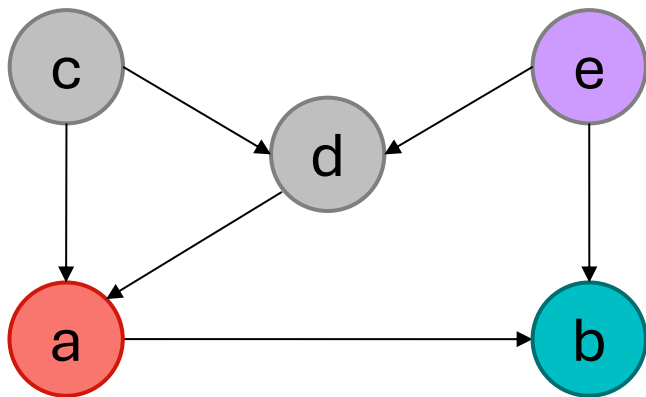
- Determine the **adjustment set** of variables to block any backdoor causal path for the estimation of the direct effect of a on b.



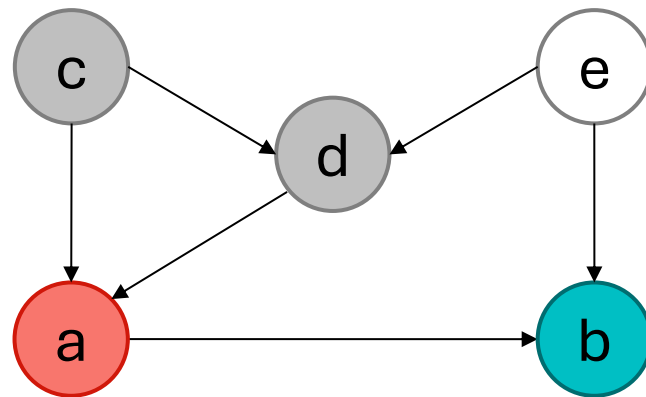
Adjustment set = {e}

Problem 3 more interesting

- Determine the **adjustment set** of variables to block any backdoor causal path for the estimation of the direct effect of a on b.

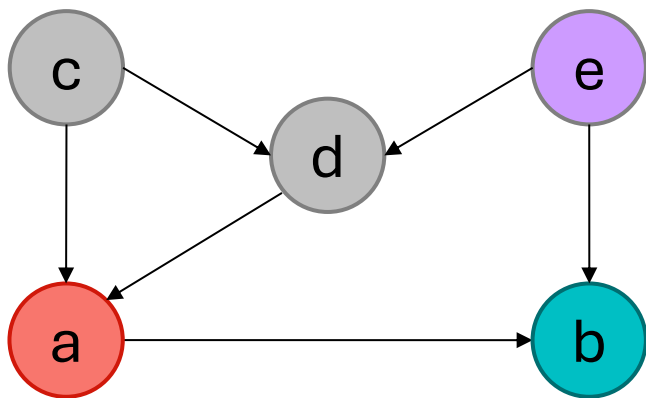


Adjustment set = {e}

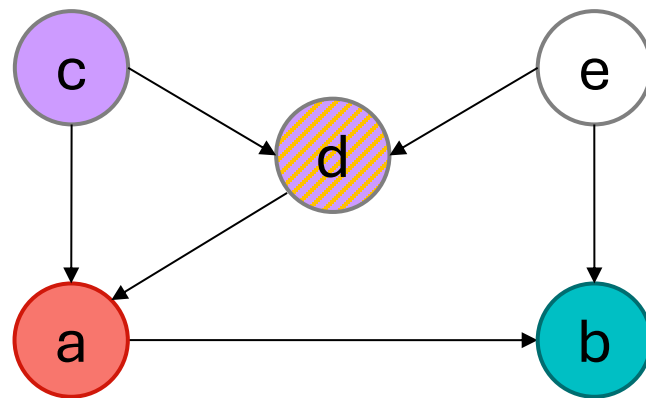


Problem 3 Solution <https://dagitty.net/m7RVrEEEnA>

- Determine the **adjustment set** of variables to block any backdoor causal path for the estimation of the direct effect of a on b.



Adjustment set = {e}



Adjustment set = {c,d}

Problem 4: Does an automated testing framework (X) reduce post-release defects (Y)?

A software quality team studies the effect of the introduction of an automated testing framework on post-release defects of a large system. They know:

1. Experienced developers are more likely to adopt the framework and to write less defect-prone code.
2. The framework lowers defects directly, and also by raising test coverage.
3. More complex modules tend to have more post-release defects.
4. Both the framework and module complexity raise the number of issues logged.

Draw a DAG including the variables: Autom. Testing, Post-Release Defects, Developer Experience, Test Coverage, Logged Issues, Code Complexity.

Identify:

- Which variables are confounders;
- Which are mediators;
- What is the minimal adjustment set to estimate **the direct and total effect** of automated testing on post-release defects.

Problem 4: Solution <https://dagitty.net/mFcJtc32s>

A software quality team studies the effect of the introduction of an automated testing framework on post-release defects of a large system. They know:

1. Experienced developers are more likely to adopt the framework and to write less defect-prone code.
2. The framework lowers defects directly, and also by raising test coverage.
3. More complex modules tend to have more post-release defects.
4. Both the framework and module complexity raise the number of issues logged.

